

Report
Of
Google AI & Machine Learning Virtual Internship

Submitted To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: 2410031218

Name of Student: DEEPAK KUMAR

Batch: 2026-27

CANDIDATE'S DECLARATION

I, Deepak Kumar, do hereby declare that the internship report titled “Google AI & Machine Learning Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science & Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Deepak Kumar

Roll No.: 2410031218

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Google AI & Machine Learning Virtual Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to Google for Developers and EduSkills, in association with AICTE, for providing this structured learning opportunity, and to the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous support and encouragement.

I also express my gratitude to my parents for their blessings.

Signature

Student Name: Deepak Kumar

Roll No.: 2410031218

INTERNSHIP COMPLETION CERTIFICATE



Certificate ID: 4de1ba3f1a15ca48ea98 | Student ID: STU69988986a6e9a1771604358

TABLE OF CONTENTS

Chapter No.	Chapter Name
1.	Candidate's Declaration
2.	Acknowledgement
3.	Internship Completion Certificate
4.	Project Description 4.1 Introduction 4.2 Organization Profile 4.3 Problem Statement 4.4 Project Objectives 4.5 Scope of the Project 4.6 Technologies and Tools Used 4.7 System Architecture 4.8 Methodology 4.9 Expected Outcomes
5.	Bibliography / References

4. Project Description

4.1 Introduction

The Google AI & Machine Learning Virtual Internship is a 10-week AICTE–EduSkills Virtual Internship Program, delivered in partnership with Google for Developers. The program is designed to build industry-relevant Artificial Intelligence and Machine Learning skills through structured, hands-on learning combined with practical project exposure. The internship was undertaken over a duration of 10 weeks, from June to August 2026, following a week-wise structured learning plan that included weekly assessments, project submissions, and a final assessment test.

4.2 Organization Profile

The internship was jointly conducted by Google for Developers and EduSkills, under the patronage of AICTE (All India Council for Technical Education), Ministry of Education, Government of India, through the National Internship Portal. Google for Developers provides curated, industry-aligned technical curricula in areas such as Artificial Intelligence, Machine Learning and TensorFlow, while EduSkills serves as the program facilitator responsible for structuring the internship, tracking learner progress, and issuing credentials upon successful completion. Together, these organizations aim to strengthen the employability of engineering students by bridging the gap between academic learning and industry-relevant technical skills.

4.3 Problem Statement

Engineering students often possess strong theoretical foundations but lack structured, hands-on exposure to real-world Artificial Intelligence and Machine Learning workflows such as data preparation, model training, evaluation, and deployment. There is a need for a guided, industry-backed learning pathway that allows students to move beyond conceptual understanding and gain practical, project-based experience in building and deploying AI/ML applications, thereby strengthening their employability and confidence in the subject area.

4.4 Project Objectives

- Understand Machine Learning fundamentals
- Learn data preparation and feature engineering

- Train and evaluate ML models
- Learn TensorFlow basics
- Understand neural networks
- Learn object detection techniques
- Apply AI concepts to real-world problems

4.5 Scope of the Project

The scope of the internship spanned four core learning modules delivered over a structured, week-wise curriculum, covering the fundamentals of neural networks, computer vision, object detection, and product image search:

- Module 1: Program Neural Networks with TensorFlow – demystifying machine learning from first principles and building convolutional neural networks (CNNs) for common computer-vision recognition and classification tasks.
- Module 2: Get Started with Object Detection – learning the basics of object detection and integrating a pretrained object detector into a mobile application.
- Module 3: Go Further with Object Detection – training custom object-detection models using TensorFlow Lite and the TensorFlow Lite Model Maker library.
- Module 4: Get Started with Product Image Search – building a product image search feature for a mobile app, and detecting objects in static images and live camera feeds on Android.

4.6 Technologies and Tools Used

The following technologies and tools were used and learned during the course of the internship:

- Python
- TensorFlow
- TensorFlow Lite
- Neural Networks
- Convolutional Neural Networks (CNN)
- Computer Vision
- Object Detection

- ML Kit
- Product Image Search

4.7 System Architecture

The end-to-end development pipeline followed during the project work is outlined below, representing the flow of data and processing stages from raw input to a deployed AI application:

**Data Collection → Data Preparation → Feature Engineering → Model Training
→ Model Evaluation → Object Detection → Deployment → AI Application**

4.8 Methodology

The internship followed a progressive, step-by-step learning path, structured as follows:

- 1. Learn AI & ML fundamentals
- 2. Understand TensorFlow
- 3. Build neural networks
- 4. Train models
- 5. Apply object detection
- 6. Deploy models
- 7. Implement product image search

4.9 Expected Outcomes

Upon successful completion of the internship, the following skills and outcomes were achieved:

- Machine Learning concepts
- TensorFlow development
- Neural networks
- Computer vision
- Object detection
- Model evaluation
- AI application development

Overall, the internship provided a strong foundation in Artificial Intelligence and Machine Learning, giving practical exposure to industry-relevant AI technologies including TensorFlow and neural networks, and a working understanding of object detection and product image search — building a solid base for future AI/ML projects.

5. Bibliography / References

Google for Developers – AI & Machine Learning Virtual Internship curriculum, <https://developers.google.com>

TensorFlow Documentation, <https://www.tensorflow.org>

TensorFlow Lite / Model Maker Documentation, <https://www.tensorflow.org/lite>

AICTE – National Internship Portal, Ministry of Education, Government of India

EduSkills Virtual Internship Program, <https://eduskillsfoundation.org>